evolves toward the fully disordered class, since there are so many page arrangements that fit this bill.

The example of *War and Peace* highlights two essential features of entropy. First, *entropy is a measure of the amount of disorder in a physical system.* High entropy means that many rearrangements of the ingredients making up the system would go unnoticed, and this in turn means the system is highly disordered (when the pages of *War and Peace* are all mixed up, any further jumbling will hardly be noticed since it simply leaves the pages in a mixed-up state). Low entropy means that very few rearrangements would go unnoticed, and this in turn means the system is highly ordered (when the pages of *War and Peace* start in their proper order, you can easily detect almost any rearrangement). Second, in physical systems with many constituents (for instance, books with many pages being tossed in the air) there is a natural evolution toward greater disorder, since disorder can be achieved in so many more ways than order. In the language of entropy, this is the statement that *physical systems tend to evolve toward states of higher entropy.*

Of course, in making the concept of entropy precise and universal, the physics definition does not involve counting the number of page rearrangements of one book or another that leave it looking the same, either ordered or disordered. Instead, the physics definition counts the number of rearrangements of fundamental constituents—atoms, subatomic particles, and so on—that leave the gross, overall, "big-picture" properties of a given physical system unchanged. As in the example of *War and Peace*, low entropy means that very few rearrangements would go unnoticed, so the system is highly ordered, while high entropy means that many rearrangements would go unnoticed, and that means the system is very disordered.[12]

For a good physics example, and one that will shortly prove handy, let's think about the bottle of Coke referred to earlier. When gas, like the carbon dioxide that was initially confined in the bottle, spreads evenly throughout a room, there are *many* rearrangements of the individual molecules that will have no noticeable effect. For example, if you flail your arms, the carbon dioxide molecules will move to and fro, rapidly changing positions and velocities. But overall, there will be no qualitative effect on their arrangement. The molecules were spread